

## Trainings ▾

### Select Service Tools

#### Recommended Cummins® Service Tools

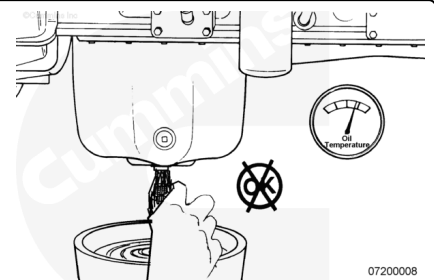
- Oil filter wrench, Part Number 3400157

#### Additional Service Items

- No additional service items required.

### Drain

Change the lubricating oil and filter at the specified oil change interval for your application. Review the Maintenance Schedule in Section 3. Refer to Procedure 359-001 in Section 3.  
(/qs3/pubsys2/xml/en/procedures/377/377-359-001.html)

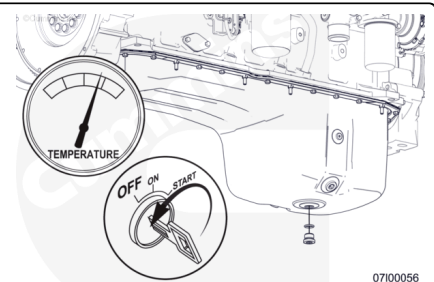


#### **⚠ WARNING ⚠**

**To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.**

#### **⚠ WARNING ⚠**

**Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.**



Operate the engine until the water temperature reaches 60°C [140°F]. Shut the engine OFF.

Remove the oil drain plug from the lubricating oil pan sump.

**Note :** Fittings used in the bottom drain opening of the oil pan other than Cummins Inc. specified parts **must not exceed** the size or weight limits, as stated in specifications for Lubricating Oil System. Refer to Procedure 018-017 in Section V.

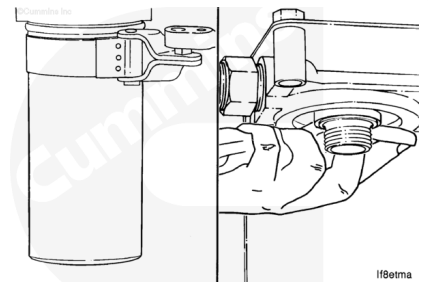
(/qs3/pubsys2/xml/en/procedures/377/377-018-017.html)

**Note :** Dispose of used oil in accordance with federal, state, and local laws and regulations.

Clean the area around the lubricating oil filter head.

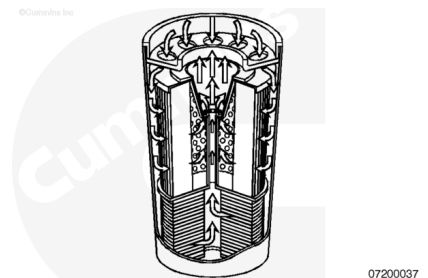
Use oil filter wrench, Part Number 3400157, to remove the lubricating oil filter.

Clean the gasket surface of the oil filter head. The o-ring can stick on the filter head. Be sure it is removed.



## Fill

Use the following procedure to locate the correct lubricating oil filter part numbers. Refer to Procedure 018-024 in Section V. (/qs3/pubsys2/xml/en/procedures/377/377-018-024.html)

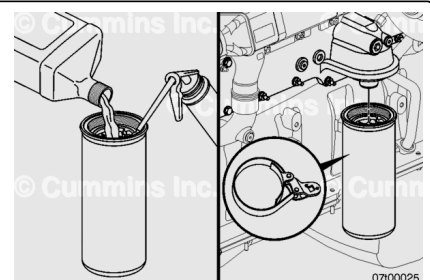


### ⚠ CAUTION ⚠

**The lack of lubrication caused by an empty oil filter is harmful to the engine.**

### ⚠ CAUTION ⚠

**Mechanical over-tightening of the filter can distort the threads or damage the filter element seal.**



Fill the lubricating oil filter with clean engine oil. Choose the correct oil for your operating climate. Refer to Procedure 018-003 in Section V. (/qs3/pubsys2/xml/en/procedures/377/377-

018-003.html) Fill the filter to the correct capacity. Use the following procedure for oil filter capacity. Refer to Procedure 018-017 in Section V.

(/qs3/pubsys2/xml/en/procedures/377/377-018-017.html)

Apply a light film of clean engine oil to the gasket sealing surface of the filter before installing the new filter.

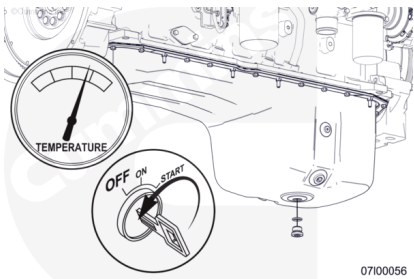
Install the filter on the filter head. Tighten the filter until the gasket contacts the filter head surface.

Use oil filter wrench, Part Number 3400157, to tighten the filter to the specifications supplied with the filter.

**Note : Check the torque of the side and bottom drain plugs to be sure both are tight.**

Clean and check the oil drain plug threads and the seal surface.

Install the oil drain plugs in both of the lubricating oil pan sumps.



**Torque Value:** 55 n·m [ 41 ft-lb ]

Use a Cummins Inc. approved, high-quality multiviscosity oil. Choose the correct oil for your operating climate. Refer to Procedure 018-003 in Section V.

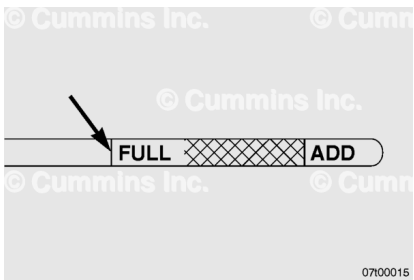
(/qs3/pubsys2/xml/en/procedures/377/377-018-003.html)

Add the specified amount of clean lubricating oil to fill the oil pan.

| Oil Pan Capacity |                        |
|------------------|------------------------|
| Stamped Steel    | 41.6 liters [11 gal]   |
| Aluminum         | 47.3 liters [12.5 gal] |

**⚠ CAUTION ⚠**

**Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut the engine OFF immediately to avoid engine damage. Confirm the correct oil level is in the oil pan.**



Fill the engine with clean oil to the FULL mark on the dipstick. Total system capacity, including the filter, is found in the following procedure. Refer to Procedure 018-017 in Section V. (/qs3/pubsys2/xml/en/procedures/377/377-018-017.html)

Operate the engine until the water temperature reaches 82°C [180°F], and check for leaks.

Shut the engine OFF. Wait approximately 15 minutes for the oil to drain back from the upper parts of the engine to the oil pan.

Check the oil level again. Add oil as necessary to bring the level up to the FULL range on the dipstick.

